Pure Mathematics with Mechanics GCE A Level 2020 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (0765 Pure Maths With Mechanics 1, printed as *"Maths With Mechanics 1"*, June 2020, Advanced Level, one and a half hours). Answer ALL questions; each question has FOUR suggested answers A, B, C and D and only one answer must be marked (if more than one answer is marked the score for that question is zero). Mathematical tables (formulae booklets) and calculators are allowed. Section A: Pure Mathematics (questions 1 to 35); Section B: Mechanics (questions 36 to 50). Mobile phones are NOT allowed in the examination room.

SECTION A: PURE MATHEMATICS

$$\frac{5x+1}{(x-1)(x+2)} \equiv$$

a) $\frac{2}{x-1} + \frac{2}{x+2}$
b) $\frac{2}{x-1} - \frac{3}{x+2}$
c) $\frac{2}{x-1} + \frac{3}{x+2}$
d) $\frac{2}{x-1} - \frac{3}{x+2}$

2. The equations of the vertical asymptotes to the graph of $f(x) = \frac{4}{x^2-1}$ are

a) $x = 1,\ x = 4$
b) $x = -1,\ x = 4$
c) $x = 1,\ x = -1$
d) $x = -1,\ x = -4$

3. Two functions f and g are defined by $f(x) = 2x + 3$ and $g(x) = x - 2$. The composite function $gf =$

a) $2x + 1$
b) $2x - 1$
c) $2x + 6$
d) $3x + 1$

4. If $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, then the cofactor of the element 6 is

Options as printed: A) $-\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$ B) $\begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$ C) $-\begin{vmatrix} -1 & -2 \\ 7 & 8 \end{vmatrix}$ D) $\begin{vmatrix} -1 & -2 \\ 7 & 8 \end{vmatrix}$

5. Given that $4^{2-x} = 4\left(2^{1-x}\right)$, the value of x is

a) $-\frac{1}{2}$
b) $\frac{1}{2}$
c) -1
d) 1

6. Given that $\ln(x+6)+\ln 9=\ln(x+7)+\ln 6$, the value of x is

a) -4
b) 4
c) -3
d) 8

7. If the roots of the quadratic equation $px^2-qx-r=0$ are real, then

a) $q^2-4pr\geq 0$
b) $q^2+4pr\geq 0$
c) $q^2+4p>0$
d) $q^2-4pr>0$

8. $\lim\limits_{n\to\infty}\left(1+\frac{2}{3n}\right)^n=$

a) $e^{3/2}$
b) $e^{2/3}$
c) e^2
d) $e^{1/3}$

9. Given two statements p and q, the converse of $p\Rightarrow q$ is

a) $p\Rightarrow\sim q$
b) $q\Rightarrow p$
c) $\sim p\Leftrightarrow q$
d) $\sim q\Rightarrow p$

10. The ordered pair that satisfies the following simultaneous equations is $2x+3y=1, 3x-y=7$.

a) $(2,1)$
b) $(-2,-1)$
c) $(1,2)$
d) $(2,-1)$

11. $\lim\limits_{x\to 0}\frac{1-\cos x}{x}=$

a) 0
b) -1
c) ∞
d) 1

12. Using the substitution $u=1+x^2$, $\int_0^1 2x(1+x^2)^3\,dx=$

a) $\frac{15}{4}$
b) $\frac{7}{4}$
c) 1
d) $\frac{17}{4}$

13. $\int \frac{2x+1}{2x}\,dx =$

a) $x + \frac{1}{2}\ln x + K$
b) $1 + \frac{1}{2}x^{-1} + K$
c) $x + 2\ln 2x + K$
d) $x + 2\ln x + K$

14. Using De Moivre's theorem, the complex number $(\cos\theta + i\sin\theta)^5$ is equal to

a) $\cos\theta^5 + i\sin\theta^5$
b) $5(\sin\theta + i\sin\theta)$
c) $\cos 5\theta + i\sin 5\theta$
d) $\cos\frac{\theta}{5} + i\sin\frac{\theta}{5}$

15. Given that $\mathbf{u}$ and $\mathbf{v}$ are coplanar vectors, where $\mathbf{u} = 2\mathbf{i} - \mathbf{j} + 6\mathbf{k}$ and $\mathbf{v} = -3\mathbf{i} + 5\mathbf{j} + \mathbf{k}$, $\mathbf{u} \times \mathbf{v} =$

a) $-31\mathbf{i} - 20\mathbf{j} + 7\mathbf{k}$
b) $-31\mathbf{i} + 20\mathbf{j} + 7\mathbf{k}$
c) $-31\mathbf{i} - 20\mathbf{j} - 7\mathbf{k}$
d) $31\mathbf{i} + 20\mathbf{j} - 7\mathbf{k}$

16. A function $f(x)$ is continuous in the interval $[a, b]$ and differentiable on the interval (a, b). The mean value theorem states that there exists a value c, where $a

a) $f'(c) = \frac{f(b) + f(a)}{a - b}$
b) $f'(c) = \frac{f(b) - f(a)}{b - a}$
c) $f'(c) = \frac{f'(a) - f'(b)}{b - a}$
d) $f'(c) = \frac{f'(b) - f'(a)}{a - b}$

17. The tangent of the acute angle between the lines $y = 4x - 3$ and $y = x - 2$ is

a) $-\frac{3}{5}$
b) $-\frac{5}{3}$
c) $\frac{3}{5}$
d) $\frac{5}{3}$

18. The general solution of the differential equation $3\frac{dy}{dx} = \frac{1}{x^2} + x$ is

a) $y = \frac{1}{3}\left(\frac{x^2}{2} - \frac{1}{x}\right) + c$
b) $y = \frac{1}{3}\left(\frac{x^2}{2} + \frac{1}{x}\right) + c$
c) $y = \frac{1}{3}\left(\frac{1}{x} - \frac{x}{2}\right) + c$
d) $y = \frac{1}{3}\left(\frac{1}{x} + \frac{x}{2}\right) + c$

19. A geometric progression has first term 16 and sum to infinity 40. The common ratio of the progression is

a) $\frac{2}{5}$
b) $\frac{3}{5}$
c) $\frac{1}{5}$
d) $\frac{4}{5}$

20. The equation of a line is $\mathbf{r} = \mathbf{i} - 3\mathbf{j} + 7\mathbf{k} + \mu(3\mathbf{i} + \mathbf{j} + 2\mathbf{k})$ and the equation of a plane is $2x + 3y - 2z = 5$. The sine of the acute angle between the line and the plane is

Options as printed: A) $\frac{2}{7}$ B) $\frac{3}{7}$ C) $\frac{1}{7}$ D) $\frac{5}{7}$

21. The solution set of $\frac{x+1}{x} \geq 0$ is

a) $\{x : x \leq -1 \text{ or } x \geq 0\}$
b) $\{x : x \leq 0 \text{ or } x \geq 1\}$
c) $\{x : -1 \leq x \leq 0\}$
d) $\{x : x \geq -1\}$

22. Given that $4\cos\theta + 3\sin\theta \equiv R\sin(\theta + \alpha)$, the value of R and $\tan\alpha$ are respectively

a) 25 and $\frac{3}{4}$
b) 5 and $\frac{3}{4}$
c) 25 and $\frac{4}{3}$
d) 5 and $\frac{4}{3}$

23. A circle with radius 4 has its centre at the point $(2, 4)$. The equation of the circle is

a) $(x-2)^2 + (y-4)^2 = 16$
b) $(x-2)^2 + (y-2)^2 = 4$
c) $(x-4)^2 + (y-2)^2 = 16$
d) $(x-2)^2 + (y-4)^2 = 4$

24. The general solution of the equation $\cos\theta = \frac{\sqrt{3}}{2}$ is

a) $\theta = 2n\pi \pm \frac{\pi}{3}$
b) $\theta = 2n\pi \pm \frac{\pi}{6}$
c) $\theta = n\pi \pm \frac{\pi}{3}$
d) $\theta = n\pi \pm \frac{\pi}{6}$

25. Given that X and Y are two variables related by the equation $Y = aX^2 + bX$, where a and b are constants. A suitable linear transformation for this non-linear equation is

a) $\frac{Y}{X} = aX + b$
b) $Y = X(aX + b)$
c) $\ln Y = \ln X + \ln(aX + b)$
d) $\ln Y = \ln(aX^2) + \ln(bX)$

26. A partial order relation is that which is

a) Reflexive, symmetric and transitive
b) Reflexive and symmetric
c) Reflexive, anti symmetric and transitive
d) Symmetric and transitive

27. Given the parametric equations $x = 2\theta + \sin 2\theta$, $y = 1 - \cos 2\theta$, where θ is the parameter, $\frac{dy}{dx} =$

a) $-\frac{\sin 2\theta}{1 - \cos 2\theta}$
b) $\frac{\cos\theta}{\sin\theta}$
c) $-\frac{\sin 2\theta}{1 + \cos 2\theta}$
d) $\frac{\sin 2\theta}{1 + \cos 2\theta}$

28. Given that $(1 + ax)^{\frac{1}{2}} \equiv 1 + \frac{x}{2} + bx^2 + \cdots$,

a) $a = -1,\ b = \frac{1}{4}$
b) $a = 1,\ b = -\frac{1}{8}$
c) $a = 1,\ b = -\frac{1}{4}$
d) $a = -1,\ b = \frac{1}{8}$

29. The number of arrangements of the letters of the word *ASSASSINATION* is

a) $\frac{13!}{3!\,4!\,2!}$
b) $\frac{13!}{4!\,3!\,2!\,2!}$
c) $\frac{5!}{4!\,3!\,4!}$
d) $\frac{5!}{4!\,3!\,2!\,2!}$

30. Given that the sum of the first n terms of a sequence is S_n, where $S_n = n(n+2)$, the fourth term of the sequence is

a) 24
b) 15
c) 11
d) 9

31. The polynomial $p(x) = x^3 + ax^2 + 2x - 5$ leaves the remainder 2 when divided by $(x-1)$. The value of a is

a) 4
b) -6
c) -6
d) 10

32. $\sum_{r=1}^{100}(r+1) =$

a) 5050
b) 200
c) 10200
d) 5150

33. A point P divides the line segment joining the points $(4,1)$ and $(5,2)$ internally in the ratio $2:3$. The coordinates of P are

a) $\left(\frac{22}{5}, 7\right)$
b) $\left(\frac{22}{5}, \frac{7}{5}\right)$
c) $\left(22, \frac{7}{5}\right)$
d) $(22, 7)$

34. A periodic function f is defined on $\mathbb{R}$, the set of real numbers, by $f(x)=\begin{cases}x+2,&0

a) 13
b) 9
c) 1
d) 3

35. The value of the constant m for which the graph of the equation $y = m(x-2)^2 + 3$ passes through the origin $(0,0)$ is

a) $\frac{3}{4}$
b) $-\frac{3}{4}$
c) $\frac{4}{3}$
d) $-\frac{4}{3}$

SECTION B: MECHANICS

At time $t = 0$ the position vectors of two particles A and B are $(5\mathbf{i} + 20\mathbf{j})$ m and $(-10\mathbf{i} - 5\mathbf{j})$ m respectively. The particles are given constant velocities $(-2\mathbf{i} - 5\mathbf{j})\ \mathrm{m\,s^{-1}}$ and $(3\mathbf{i} + 5\mathbf{j})\ \mathrm{m\,s^{-1}}$ respectively. The position vector of B relative to the position vector of A after 3 s is

a) $5\mathbf{j}$ m
b) $-5\mathbf{j}$ m
c) $(-15\mathbf{i} - 25\mathbf{j})$ m
d) $(5\mathbf{i} + 10\mathbf{j})$ m

37. Two particles of mass 2 kg and 3 kg are placed at the points $(3, 1)$ and $(8, 6)$ respectively. The position of their centre of mass is at the point

a) $(4, 6)$
b) $(5.5, 3.5)$
c) $(5, 3)$
d) $(6, 4)$

38. A particle is projected with speed $20\ \mathrm{m\,s^{-1}}$ at an angle 30° to the horizontal through the point of projection. Taking g as $10\ \mathrm{m\,s^{-2}}$, the greatest height reached by the particle from the point of projection is

a) 25 m
b) 5 m
c) $20\sqrt{3}$ m
d) 10 m

39. A lorry of mass 1000 kg is moving with speed $3\ \mathrm{m\,s^{-1}}$ when a 200 kg sack of rubbish is dropped vertically into the lorry. The new speed of the lorry is

a) $12\ \mathrm{m\,s^{-1}}$
b) $2\frac{1}{2}\ \mathrm{m\,s^{-1}}$
c) $2\frac{6}{7}\ \mathrm{m\,s^{-1}}$
d) $\dfrac{3}{4}\ \mathrm{m\,s^{-1}}$

40. A body of weight 20 N is suspended from a fixed point O by a light elastic string of natural length 0.3 m and modulus of elasticity 200 N. The extension of the string is

a) 0.03 m
b) 0.3 m
c) 3 m
d) 0.12 m

41. A particle is projected from a point on a horizontal plane with speed V at an angle θ to the plane. The horizontal range R covered by the particle is

a) $R = \dfrac{2V^2 \sin 2\theta}{g}$
b) $R = \dfrac{V^2 \cos\theta \sin\theta}{g}$
c) $R = \dfrac{V^2 \sin 2\theta}{2g}$
d) $R = \dfrac{V^2 \sin 2\theta}{g}$

42. Two forces $\mathbf{R} = (2\mathbf{i} + 3\mathbf{j})$ N and $\mathbf{F} = (8\mathbf{i} + 7\mathbf{j})$ N act on a body of mass 10 kg. The velocity of the body 4 s after starting from rest is

a) $(24\mathbf{i} + 16\mathbf{j})\ \mathrm{m\,s^{-1}}$
b) $(4.0\mathbf{i} + 4.0\mathbf{j})\ \mathrm{m\,s^{-1}}$
c) $(2.4\mathbf{i} + 1.6\mathbf{j})\ \mathrm{m\,s^{-1}}$
d) $(40\mathbf{i} + 40\mathbf{j})\ \mathrm{m\,s^{-1}}$

43. A force of 14 N acts on a particle of mass 70 kg causing the speed of the particle to increase from $3\ \mathrm{m\,s^{-1}}$ to $7\ \mathrm{m\,s^{-1}}$. The distance travelled by the particle during this period is

a) 40 m
b) 4 m
c) 100 m
d) 10 m

44. Fig. 1 shows two particles P and Q hanging freely and connected by a light inextensible string passing over a fixed smooth pulley.

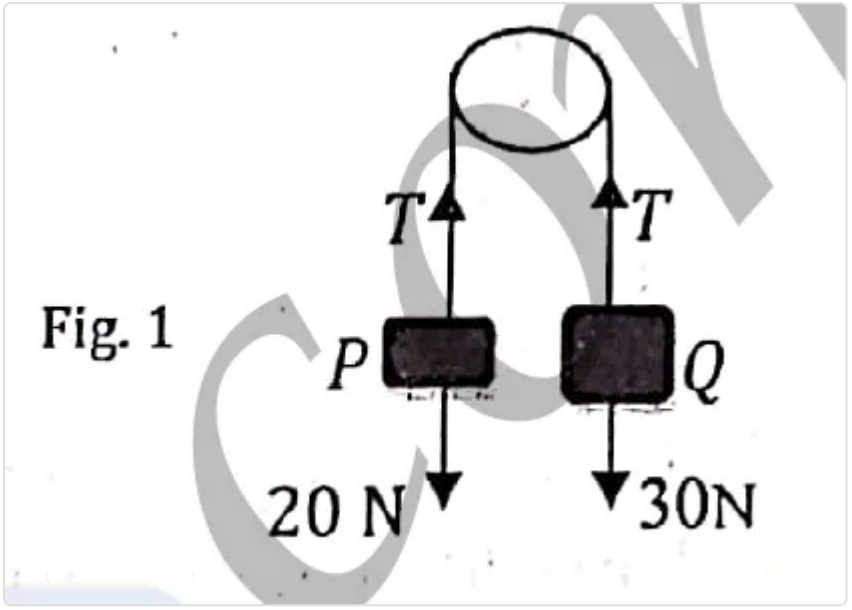


Fig. 1

(Fig. 1: P has weight 20 N, Q has weight 30 N, and the tension T acts upwards on each particle.)

The system is released from rest. Taking g as $10\ \mathrm{m\,s^{-2}}$, the acceleration of P and the tension in the string are

a) $10\ \mathrm{m\,s^{-2}}$ and 24 N
b) $2\ \mathrm{m\,s^{-2}}$ and 20 N
c) $10\ \mathrm{m\,s^{-2}}$ and 30 N
d) $2\ \mathrm{m\,s^{-2}}$ and 24 N

45. A particle of mass 4 kg is moving in a horizontal circle of radius 2 m. The centripetal force is 72 N. The angular speed of the particle is

a) $9\ \mathrm{rad\,s^{-1}}$
b) $3\ \mathrm{rad\,s^{-1}}$
c) $6\ \mathrm{m\,s^{-1}}$
d) $3\ \mathrm{m\,s^{-1}}$

46. The position vector of a particle is $\mathbf{r} = \left[(t^3 + 2t^2 + 2)\mathbf{i} + (t^2 - 1)\mathbf{j}\right]$ m at time t s. The velocity of the particle when $t = 1$ is

a) $(9\mathbf{i} + \mathbf{j})\ \mathrm{m\,s^{-1}}$
b) $(7\mathbf{i} + 2\mathbf{j})\ \mathrm{m\,s^{-1}}$
c) $(7\mathbf{i} + \mathbf{j})\ \mathrm{m\,s^{-1}}$
d) $5\mathbf{i}\ \mathrm{m\,s^{-1}}$

47. The speeds of a car during 8 s of braking are given in the table below.

Time, t s	**0**	**2**	**4**	**6**	**8**
Speed, v m s^{-1}	30	27	22	14	0

Using the trapezium rule, the distance covered by the car before coming to rest is approximately

a) 156 m
b) 158 m
c) 157 m
d) 155 m

48. The engine of a car of mass 1200 kg works at a constant rate of 16 kW. The car is moving on a straight level road with constant resistance of magnitude 400 N to its motion. The maximum speed of the car is

a) $40\ \mathrm{m\,s^{-1}}$
b) $4\ \mathrm{m\,s^{-1}}$
c) $13\ \mathrm{m\,s^{-1}}$
d) $80\ \mathrm{m\,s^{-1}}$

49. A fair die is thrown two times in succession. The probability of throwing at least one six is

a) $\frac{1}{36}$
b) $\frac{12}{36}$
c) $\frac{11}{36}$
d) $\frac{2}{6}$

50. The work done in stretching an elastic string of natural length 5 m and modulus of elasticity 500 N from a total length of 6 m to 7 m is

a) 1300 J
b) 400 J
c) 150 J
d) 300 J

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